

SafeTransit

Safety Navigation App

PROJECT TYPE Safety Navigation App	PLATFORM Cross-platform mobile application
AUDIENCE Women commuters and trusted family members	PRIMARY FOCUS Safety-first urban navigation

Overview

A safety-first navigation mobile app designed for women commuters in Philippine cities. Built with React Native and Expo SDK 54, SafeTransit pairs intelligent multi-modal route planning with community-powered safety alerts, family location sharing, and a discreet Silent Panic Button to create a complete journey protection network.

Core Features

- Background Protection: continuous location monitoring with danger-zone alerts even when the app is closed
- Silent Panic Button that fires a discreet alert to helpers and emergency contacts
- Quick Exit Screen that instantly disguises the app as a weather forecast for privacy
- Multi-modal Safe Route Planning (Walking, Driving, Transit) with color-coded safety scoring
- Community map tips with admin verification, severity color coding, and intelligent clustering
- Community forum with upvotes, threaded comments, flairs, and photo attachments via UploadThing
- Real-time family location sharing with trusted members and live map indicators
- JWT auth with Google Sign-In, token refresh, and secure bearer-token API communication

Experience and Design

- A safety heatmap and verified community tips provide context before a trip begins.
- Route planning compares walking, driving, and transit options using safety scoring.
- Family sharing communicates live journey status with selected trusted contacts.
- A Silent Panic Button and Quick Exit Screen support discreet action during unsafe situations.

Architecture and Data

- React Native and Expo provide the mobile interface, navigation, and device integrations.
- NativeWind supplies shared responsive styling across supported mobile platforms.
- Vercel-hosted APIs and Neon PostgreSQL store accounts, tips, forum content, and relationships.
- Map, routing, geocoding, and upload services are isolated behind service-layer modules.

Security and Privacy

- JWT authentication and Google Sign-In protect account-bound safety information.
- Trusted-contact relationships control who can view shared family locations.
- Location and emergency data are transmitted only through authenticated requests.
- Upload validation and server-side authorization protect community images and forum content.

Technology Stack

React Native 0.81	Application framework and UI
Expo SDK 54	Application capability
TypeScript	Application language
NativeWind	Interface styling and design system
React Native Maps	Application framework and UI
Neon PostgreSQL	Authentication, database, or cloud data
Vercel	Application hosting
UploadThing	Media storage and delivery

Performance and Reliability

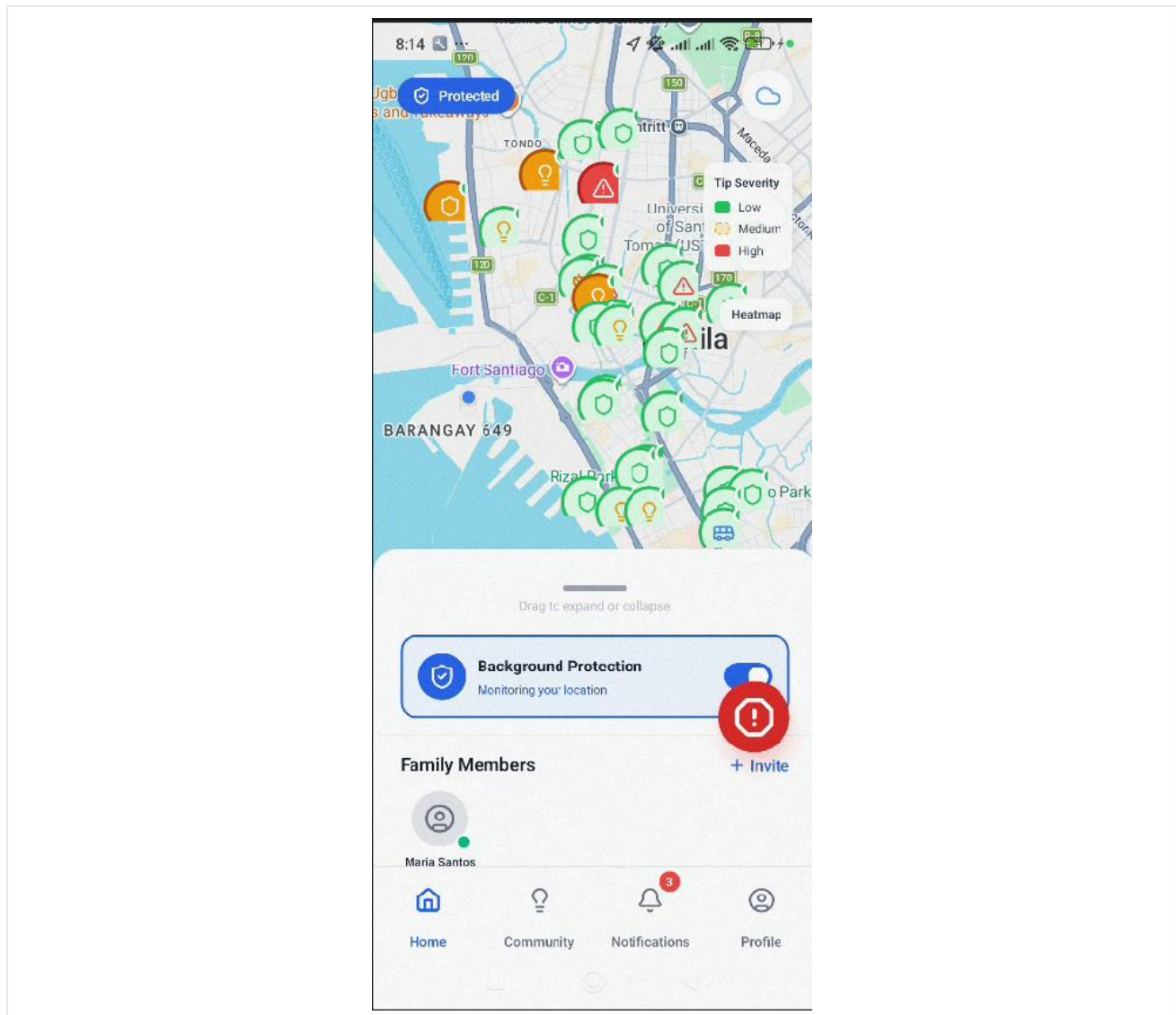
- Map clustering limits marker overload in dense urban areas.
- Background location monitoring is scoped to active protection scenarios.
- Cached map and community data reduce repeated requests during navigation.
- Reanimated and gesture handling keep map interactions smooth on mobile devices.

Platform Requirements

- Supported Android or iOS device with location services enabled.
- Internet connection for routing, safety tips, forums, and family sharing.
- Location permission for heatmaps, route planning, and background protection.
- Notification permission is recommended for alerts and trusted-contact updates.

PAGE OVERVIEW

Home - Safety Heatmap

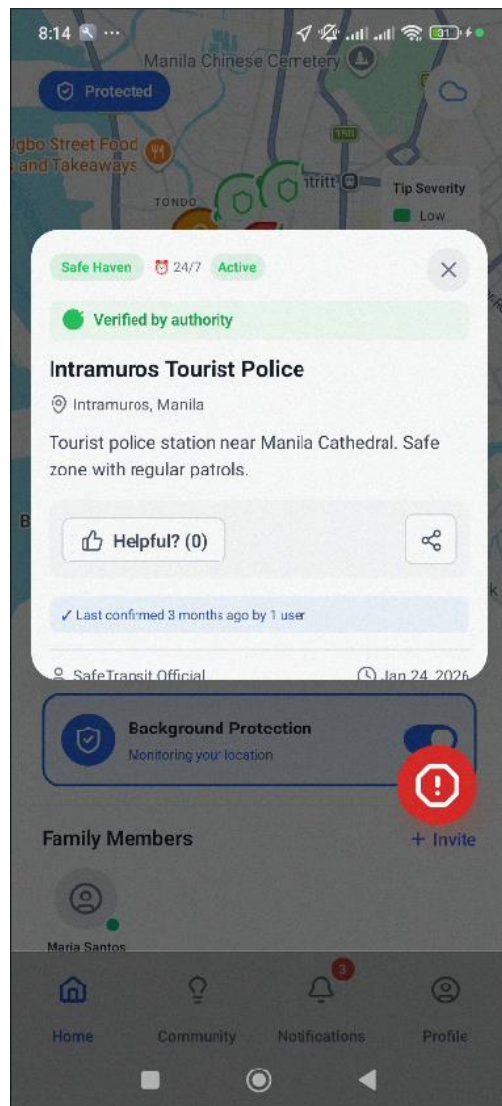


Home - Safety Heatmap

Interface view from SafeTransit.

PAGE OVERVIEW

Tip Detail

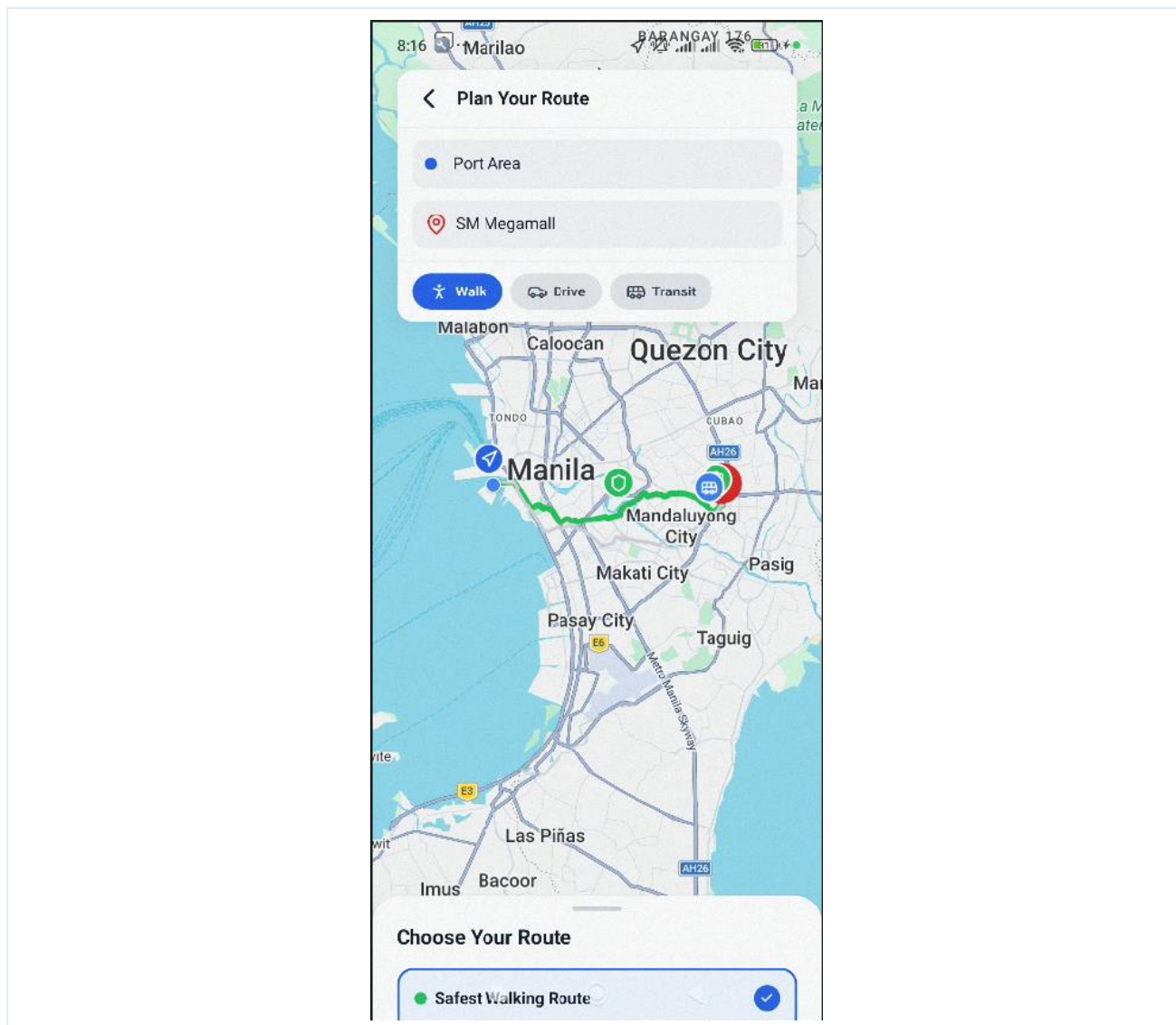


Tip Detail

Interface view from SafeTransit.

PAGE OVERVIEW

Safe Route Planner

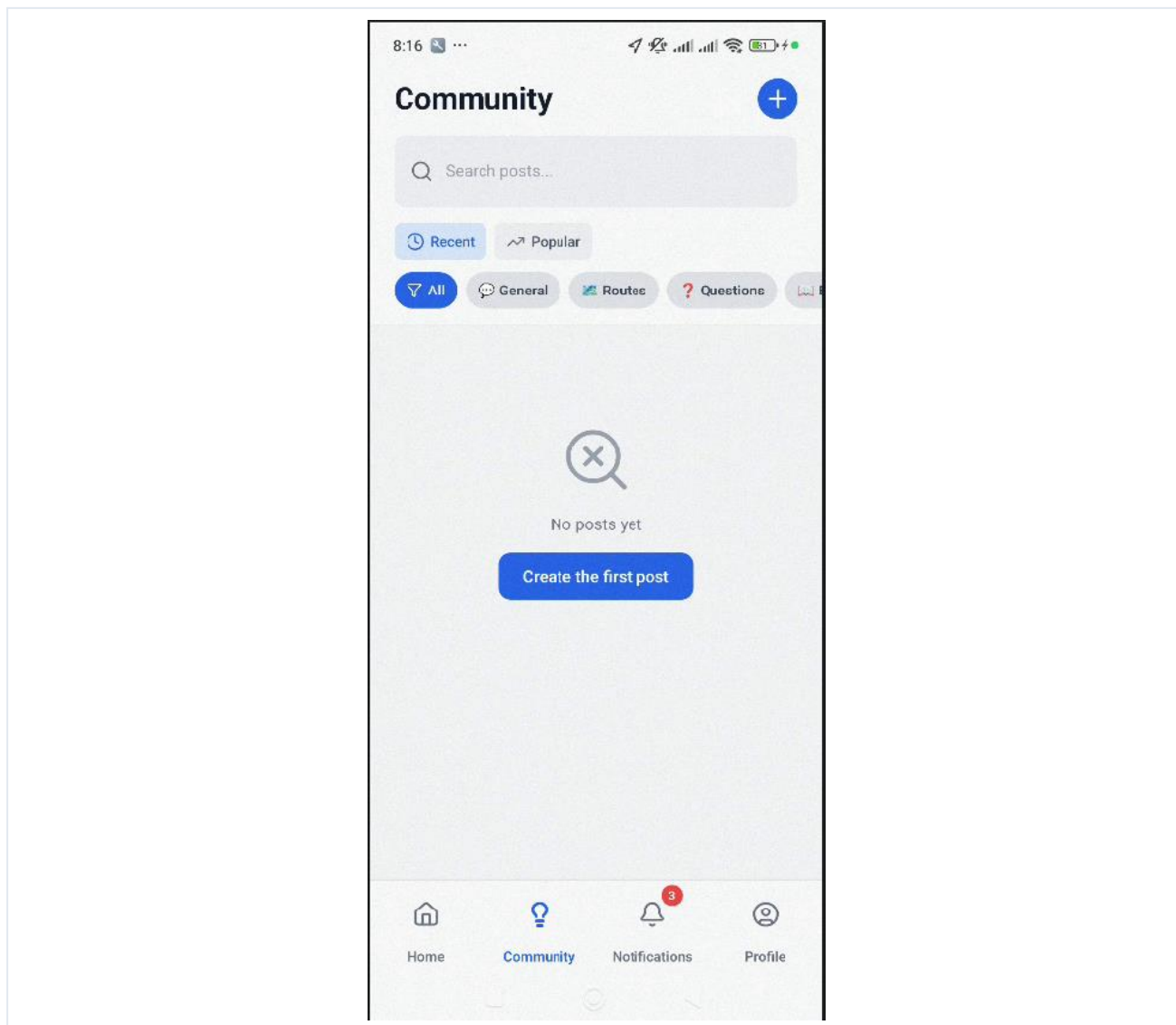


Safe Route Planner

Interface view from SafeTransit.

PAGE OVERVIEW

Community Forum

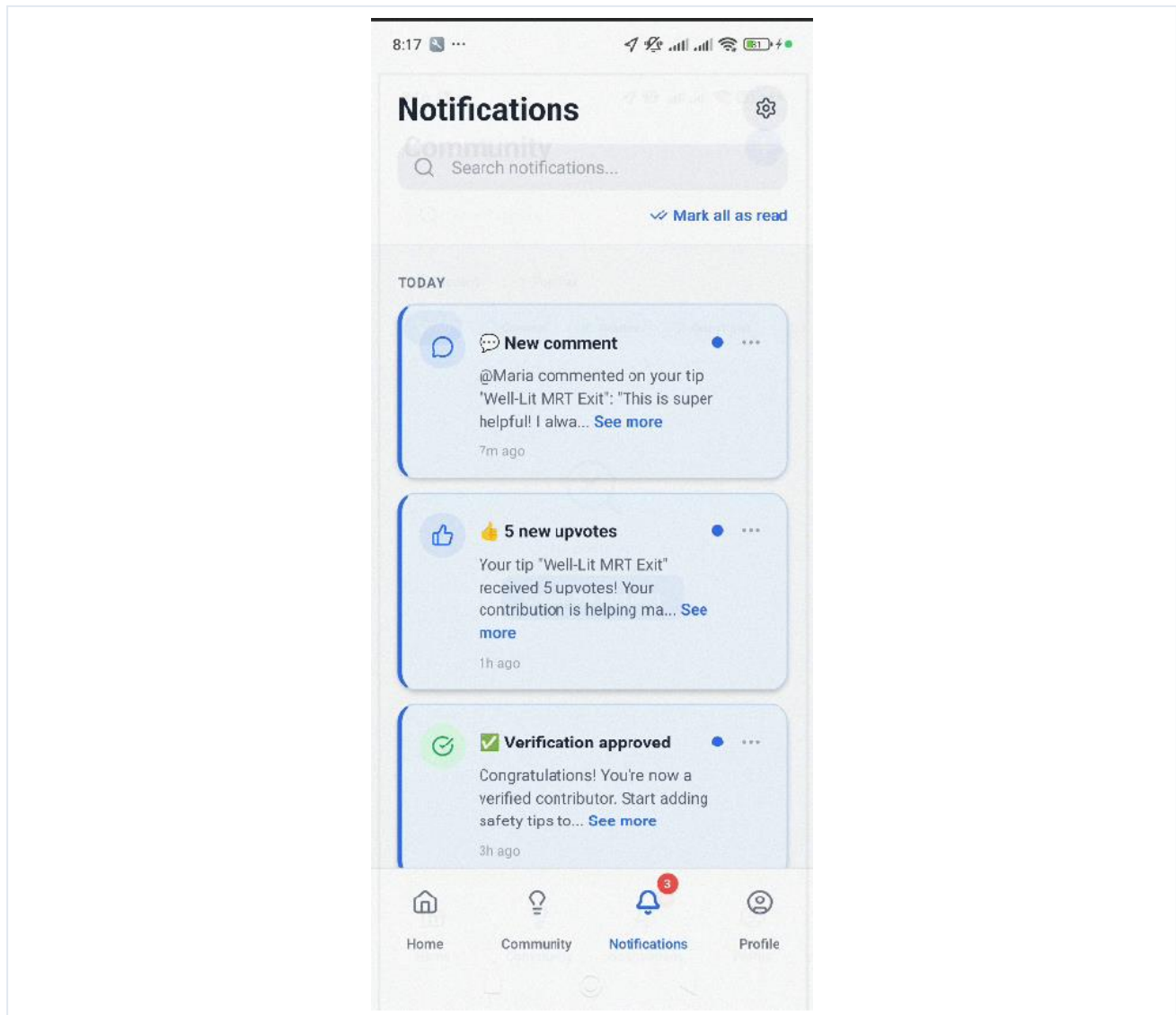


Community Forum

Interface view from SafeTransit.

PAGE OVERVIEW

Notifications

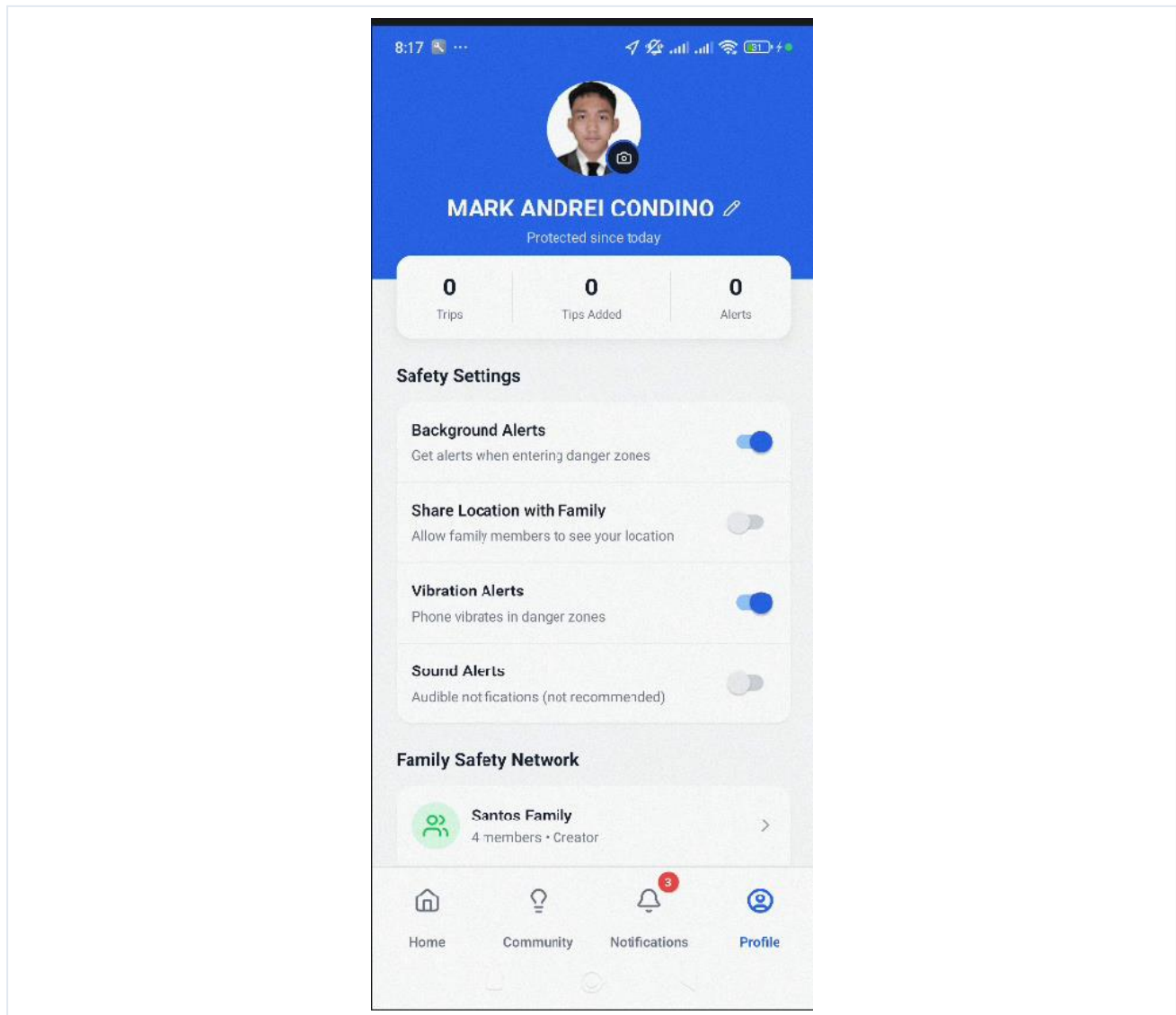


Notifications

Interface view from SafeTransit.

PAGE OVERVIEW

Profile & Safety Settings

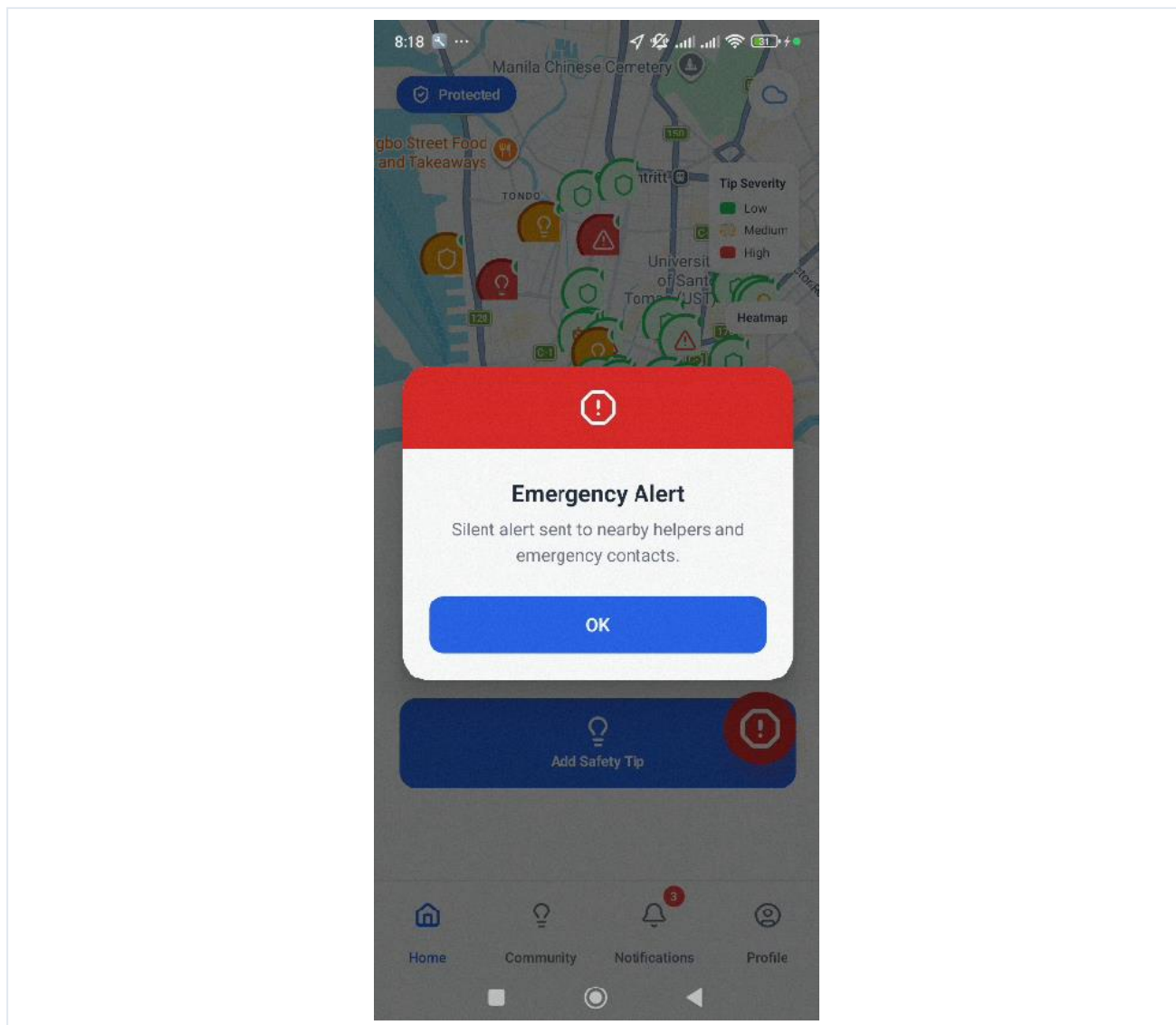


Profile & Safety Settings

Interface view from SafeTransit.

PAGE OVERVIEW

Silent Panic Button



Silent Panic Button

Interface view from SafeTransit.